

WIRESHARK PACKET ANALYSIS REPORT

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1. ABSTRACT

Network traffic analysis is an important aspect of cybersecurity and network administration. Wireshark is a widely used network protocol analyzer that allows users to capture and inspect packets traveling across a network. This project focuses on capturing live network traffic and analyzing common protocols such as DNS, HTTP, and TCP. The captured packets were examined to understand network communication and identify any unusual or suspicious traffic patterns.

2. OBJECTIVE

The objective of this project is:

- To capture live network traffic using Wireshark.
- To analyze DNS, HTTP, and TCP packets.
- To understand the TCP three-way handshake process.
- To identify suspicious or abnormal network packets.
- To gain practical experience in network packet analysis.

3. TOOLS USED

Tool | Purpose

Wireshark | Network Packet Capture and Analysis

Windows Operating System | Platform for Analysis

Internet Connection | Generation of Network Traffic

4. METHODOLOGY

Step 1: Installation of Wireshark

Wireshark was downloaded and installed successfully on the system. The Npcap driver was installed during setup to enable packet capturing from the network interface.

Step 2: Capturing Network Traffic

The Wi-Fi interface was selected in Wireshark and packet capture was started. Various internet activities such as browsing websites and accessing online services were performed to generate network traffic.

Step 3: DNS Packet Analysis

A DNS filter was applied using:

dns

The DNS packets were analyzed to observe how domain names are resolved into IP addresses.

Figure 1: DNS Packet Analysis

The image shows a Wireshark packet capture of DNS traffic. The top pane displays a list of 23 DNS packets. The bottom pane shows the details of the selected packet (No. 7), which is a Standard query for grammarly.com.

No.	Time	Source	Destination	Protocol	Length	Info
7	0.191222800	10.244.227.10	10.244.227.127	DNS	76	Standard query 0x8c97 A in.grammarly.com
8	0.191803800	10.244.227.10	10.244.227.127	DNS	76	Standard query 0x845a AAAA in.grammarly.com
9	0.227191200	2409:40f4:1010:a141...	2409:40f4:1010:a141...	DNS	96	Standard query 0x8c97 A in.grammarly.com
10	0.227594000	2409:40f4:1010:a141...	2409:40f4:1010:a141...	DNS	96	Standard query 0x845a AAAA in.grammarly.com
11	0.230659700	10.244.227.127	10.244.227.10	DNS	172	Standard query response 0x8c97 A in.grammarly.com A
12	0.230823800	10.244.227.127	10.244.227.10	DNS	244	Standard query response 0x845a AAAA in.grammarly.co
13	0.232112700	2409:40f4:1010:a141...	2409:40f4:1010:a141...	DNS	192	Standard query response 0x8c97 A in.grammarly.com A
14	0.232114300	2409:40f4:1010:a141...	2409:40f4:1010:a141...	DNS	264	Standard query response 0x845a AAAA in.grammarly.co
16	0.377586700	10.244.227.10	10.244.227.127	DNS	78	Standard query 0x1cf7 A gnar.grammarly.com
17	0.378048900	10.244.227.10	10.244.227.127	DNS	78	Standard query 0x12d1 AAAA gnar.grammarly.com
18	0.411042500	10.244.227.127	10.244.227.10	DNS	206	Standard query response 0x1cf7 A gnar.grammarly.com
19	0.412370300	10.244.227.127	10.244.227.10	DNS	302	Standard query response 0x12d1 AAAA gnar.grammarly.
51	1.350636700	10.244.227.10	10.244.227.127	DNS	81	Standard query 0x872a A gateway.grammarly.com
52	1.351310600	10.244.227.10	10.244.227.127	DNS	81	Standard query 0x43d9 AAAA gateway.grammarly.com
53	1.379394300	10.244.227.127	10.244.227.10	DNS	209	Standard query response 0x872a A gateway.grammarly.

Frame 7: Packet, 76 bytes on wire (608 bits), 76 bytes captured (608 bits) on interface 0	Time	Source	Destination	Protocol	Length	Info
7	0.191222800	10.244.227.10	10.244.227.127	DNS	76	Standard query 0x8c97 A in.grammarly.com

Details of Frame 7:

- Ethernet II, Src: LiteonTechno 1e:11:3f (74:e5:43:1e:11:3f), Dst: 08:00:00:00:00:00
- Internet Protocol Version 4, Src: 10.244.227.10, Dst: 10.244.227.127
- User Datagram Protocol, Src Port: 51098, Dst Port: 53
- Domain Name System (query)
 - Standard query query 0x8c97 A in.grammarly.com

Observation:

The captured DNS packets show query and response messages for domain resolution. DNS requests were generated when accessing internet resources, and the DNS server responded with the corresponding IP addresses.

Step 4: HTTP Packet Analysis

An HTTP filter was applied using:

http

HTTP packets were analyzed to observe communication between the client and web server.

Figure 2: HTTP Packet Analysis

No.	Time	Source	Destination	Protocol	Length	Info
4	0.003269600	2600:1900:4110:86f::	2409:40f4:1010:a141::	HTTP	96	HTTP/1.1 206 Partial Content (application/vnd.google.octet-stream-compressible)
41	1.193225000	2409:40f4:1010:a141::	2600:1900:4110:86f::	HTTP	448	GET /edgedl/release2/chrome_component/acddvvyvhts76ngei465tcu7besa_4.10.3050.0/oimompecagnajdejgnnjjobeb...
49	1.242342300	2600:1900:4110:86f::	2409:40f4:1010:a141::	HTTP	216	HTTP/1.1 206 Partial Content (application/vnd.google.octet-stream-compressible)
79	2.287679600	2409:40f4:1010:a141::	2600:1900:4110:86f::	HTTP	449	GET /edgedl/release2/chrome_component/acddvvyvhts76ngei465tcu7besa_4.10.3050.0/oimompecagnajdejgnnjjobeb...
89	2.337614400	2600:1900:4110:86f::	2409:40f4:1010:a141::	HTTP	1125	HTTP/1.1 206 Partial Content (application/vnd.google.octet-stream-compressible)
110	3.501618600	2409:40f4:1010:a141::	2600:1900:4110:86f::	HTTP	450	GET /edgedl/release2/chrome_component/acddvvyvhts76ngei465tcu7besa_4.10.3050.0/oimompecagnajdejgnnjjobeb...
121	3.568756500	2600:1900:4110:86f::	2409:40f4:1010:a141::	HTTP	1003	HTTP/1.1 206 Partial Content (application/vnd.google.octet-stream-compressible)
154	4.591068500	2409:40f4:1010:a141::	2600:1900:4110:86f::	HTTP	450	GET /edgedl/release2/chrome_component/acddvvyvhts76ngei465tcu7besa_4.10.3050.0/oimompecagnajdejgnnjjobeb...
180	4.804437100	2600:1900:4110:86f::	2409:40f4:1010:a141::	HTTP	490	HTTP/1.1 206 Partial Content (application/vnd.google.octet-stream-compressible)
199	5.778256700	2409:40f4:1010:a141::	2600:1900:4110:86f::	HTTP	450	GET /edgedl/release2/chrome_component/acddvvyvhts76ngei465tcu7besa_4.10.3050.0/oimompecagnajdejgnnjjobeb...
244	5.850387800	2600:1900:4110:86f::	2409:40f4:1010:a141::	HTTP	814	HTTP/1.1 206 Partial Content (application/vnd.google.octet-stream-compressible)
292	8.557144900	2409:40f4:1010:a141::	2600:1900:4110:86f::	HTTP	451	GET /edgedl/release2/chrome_component/acddvvyvhts76ngei465tcu7besa_4.10.3050.0/oimompecagnajdejgnnjjobeb...
384	8.684045400	2600:1900:4110:86f::	2409:40f4:1010:a141::	HTTP	134	HTTP/1.1 206 Partial Content (application/vnd.google.octet-stream-compressible)
422	9.606217200	2409:40f4:1010:a141::	2600:1900:4110:86f::	HTTP	452	GET /edgedl/release2/chrome_component/acddvvyvhts76ngei465tcu7besa_4.10.3050.0/oimompecagnajdejgnnjjobeb...
624	9.717222200	2600:1900:4110:86f::	2409:40f4:1010:a141::	HTTP	1178	HTTP/1.1 206 Partial Content (application/vnd.google.octet-stream-compressible)
656	10.636038800	2409:40f4:1010:a141::	2600:1900:4110:86f::	HTTP	452	GET /edgedl/release2/chrome_component/acddvvyvhts76ngei465tcu7besa_4.10.3050.0/oimompecagnajdejgnnjjobeb...
1028	10.761931400	2600:1900:4110:86f::	2409:40f4:1010:a141::	HTTP	725	HTTP/1.1 206 Partial Content (application/vnd.google.octet-stream-compressible)
1097	11.658507000	2409:40f4:1010:a141::	2600:1900:4110:86f::	HTTP	453	GET /edgedl/release2/chrome_component/acddvvyvhts76ngei465tcu7besa_4.10.3050.0/oimompecagnajdejgnnjjobeb...

Frame 4: Packet, 96 bytes on wire (768 bits), 96 bytes captured (768 bits) on interface \Device...	0000	74 e5 43 1e 11 3f 46 42 36 57 64 39 86 dd 60 09	t C ?FB 6Md9
Ethernet II, Src: 46:42:36:57:64:39 (46:42:36:57:64:39), Dst: LiteonTechno 1e:11:3f (74:e5:43:1e:11:3f)	0010	2d e7 00 2a 06 3b 26 00 19 00 41 10 08 6f 00 00	...:8 ...A ...o...
Internet Protocol Version 6, Src: 2600:1900:4110:86f::, Dst: 2409:40f4:1010:a141:69ff:ec07:2ac	0020	00 00 00 00 00 00 24 09 40 f4 10 10 a1 41 69 ff	\$ @...A...
Transmission Control Protocol, Src Port: 80, Dst Port: 3656, Seq: 2441, Ack: 1, Len: 22	0030	ec 07 2a db 96 07 00 50 0e 48 36 80 72 64 8c 34	...*...P H6 rd 4
[3 Reassembled TCP Segments (2462 bytes): #2(1220), #3(1220), #4(22)]	0040	1b 64 58 18 04 1a 22 4f 00 00 00 b8 75 00 40 01	..dP...O ...u @
Hypertext Transfer Protocol	0050	00 00 00 00 00 00 00 00 00 00 60 bf 56 01 08 06	v...
Media Type			

Observation:

HTTP packets were successfully captured. The traffic includes HTTP requests and server responses used for web communication. The analysis demonstrates how web resources are requested and delivered over the network.

Step 5: TCP Packet Analysis

A TCP filter was applied using:

tcp

TCP packets were analyzed to understand connection-oriented communication.

Figure 3: TCP Packet Analysis

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	2600:1900:4110:86f::	2409:40f4:1010:a141::	TCP	74	80 → 3656 [ACK] Seq=1 Ack=1 Win=1050 Len=0
2	0.003115200	2600:1900:4110:86f::	2409:40f4:1010:a141::	TCP	1294	80 → 3656 [ACK] Seq=1 Ack=1 Win=1050 Len=1220 [TCP PDU reassembled in 4]
3	0.003289600	2600:1900:4110:86f::	2409:40f4:1010:a141::	TCP	1294	80 → 3656 [PSH, ACK] Seq=1221 Ack=1 Win=1050 Len=1220 [TCP PDU reassembled in 4]
4	0.003289600	2600:1900:4110:86f::	2409:40f4:1010:a141::	HTTP	96	HTTP/1.1 200 Partial Content (application/vnd.google.protobuf-stream-compressible)
5	0.003349400	2409:40f4:1010:a141::	2600:1900:4110:86f::	TCP	74	3656 → 80 [ACK] Seq=1 Ack=2463 Win=512 Len=0
6	0.007420900	2409:40f4:1010:a141::	2405:200:1630:a03::	TCP	74	3610 → 443 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
15	0.232179200	2409:40f4:1010:a141::	64:ff9b::6259:bac8	TCP	86	3657 → 443 [SYN] Seq=0 Win=64320 Len=0 MSS=1340 WS=256 SACK_PERM
20	0.413322900	2409:40f4:1010:a141::	64:ff9b::22c6:2f79	TCP	86	3658 → 443 [SYN] Seq=0 Win=64320 Len=0 MSS=1340 WS=256 SACK_PERM
21	0.488348100	64:ff9b::6259:bac8	2409:40f4:1010:a141::	TCP	86	443 → 3657 [SYN, ACK] Seq=0 Ack=1 Win=26883 Len=0 MSS=1300 SACK_PERM WS=256
22	0.488470100	2409:40f4:1010:a141::	64:ff9b::6259:bac8	TCP	74	3657 → 443 [ACK] Seq=1 Ack=1 Win=131072 Len=0
23	0.488895300	2409:40f4:1010:a141::	64:ff9b::6259:bac8	TLSv1.2	422	Client Hello (SNI=in.grammarly.com)
24	0.787824000	64:ff9b::6259:bac8	2409:40f4:1010:a141::	TCP	76	443 → 3657 [ACK] Seq=1 Ack=349 Win=28160 Len=0
25	0.787971300	64:ff9b::6259:bac8	2409:40f4:1010:a141::	TLSv1.2	1354	Server Hello
26	0.788120500	64:ff9b::6259:bac8	2409:40f4:1010:a141::	TCP	1354	443 → 3657 [ACK] Seq=1281 Ack=349 Win=28160 Len=1280 [TCP PDU reassembled in 29]
27	0.788170000	2409:40f4:1010:a141::	64:ff9b::6259:bac8	TCP	74	3657 → 443 [ACK] Seq=349 Ack=2561 Win=131072 Len=0
28	0.788383800	64:ff9b::6259:bac8	2409:40f4:1010:a141::	TCP	1354	443 → 3657 [ACK] Seq=2561 Ack=349 Win=28160 Len=1280 [TCP PDU reassembled in 29]
29	0.788385500	64:ff9b::6259:bac8	2409:40f4:1010:a141::	TLSv1.2	449	Certificate, Server Key Exchange, Server Hello Done
30	0.788386800	64:ff9b::22c6:2f79	2409:40f4:1010:a141::	TCP	86	443 → 3658 [SYN, ACK] Seq=0 Ack=1 Win=26883 Len=0 MSS=1300 SACK_PERM WS=256

Observation:

TCP packets containing ACK, SYN, and other control flags were observed. These packets demonstrate reliable communication between client and server devices.

Step 6: Suspicious Packet Analysis

The captured traffic was inspected for abnormal or unusual packets.

Figure 4: Suspicious Packet Analysis

The image shows a Wireshark packet capture window. The main pane displays a list of network packets. Packet 6, at time 0.007420900, is a TCP segment from 2409:40f4:1010:a141 to 2405:200:1630:a03. It is an RST (Reset) packet with the ACK flag set and a sequence number of 443. This packet is highlighted in red. The packet details pane on the right shows the structure of this packet: Ethernet II, Internet Protocol Version 6, Transmission Control Protocol (RST), and Hypertext Transfer Protocol. The packet bytes pane shows the raw data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	2600:1900:4110:86f::	2409:40f4:1010:a141...	TCP	74	80 → 3656 [ACK] Seq=1 Ack=1 Win=1050 Len=0
2	0.003115200	2600:1900:4110:86f::	2409:40f4:1010:a141...	TCP	1294	80 → 3656 [ACK] Seq=1 Ack=1 Win=1050 Len=1220 [TCP PDU reassembled in 4]
3	0.003288200	2600:1900:4110:86f::	2409:40f4:1010:a141...	TCP	1294	80 → 3656 [PSH, ACK] Seq=1221 Ack=1 Win=1050 Len=1220 [TCP PDU reassembled in 4]
4	0.003289600	2600:1900:4110:86f::	2409:40f4:1010:a141...	HTTP	96	HTTP/1.1 200 Partial Content (application/vnd.google.octet-stream-compressible)
5	0.003349400	2409:40f4:1010:a141...	2600:1900:4110:86f::	TCP	74	3656 → 80 [ACK] Seq=1 Ack=2463 Win=512 Len=0
6	0.007420900	2409:40f4:1010:a141...	2405:200:1630:a03::	TCP	74	3610 → 443 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
7	0.191222000	10.244.227.10	10.244.227.127	DNS	76	Standard query 0x8c97 A in.grammarly.com
8	0.191983000	10.244.227.10	10.244.227.127	DNS	76	Standard query 0x845a AAAA in.grammarly.com
9	0.227191200	2409:40f4:1010:a141...	2409:40f4:1010:a141...	DNS	96	Standard query 0x8c97 A in.grammarly.com
10	0.227594000	2409:40f4:1010:a141...	2409:40f4:1010:a141...	DNS	96	Standard query 0x845a AAAA in.grammarly.com
11	0.230659700	10.244.227.127	10.244.227.10	DNS	172	Standard query response 0x8c97 A in.grammarly.com A 52.202.177.29 A 3.222.246.204 A 34.196.167.243 A 18.2...
12	0.230823000	10.244.227.127	10.244.227.10	DNS	244	Standard query response 0x845a AAAA in.grammarly.com AAAA 64:ff9b:6259:bac8 AAAA 64:ff9b:36d3:240e AAAA...
13	0.232112700	2409:40f4:1010:a141...	2409:40f4:1010:a141...	DNS	192	Standard query response 0x8c97 A in.grammarly.com A 54.211.36.14 A 98.89.186.200 A 18.232.194.142 A 34.19...
14	0.232114300	2409:40f4:1010:a141...	2409:40f4:1010:a141...	DNS	264	Standard query response 0x845a AAAA in.grammarly.com AAAA 64:ff9b:12e8:c28e AAAA 64:ff9b:22c4:a7f3 AAAA...
15	0.232179200	2409:40f4:1010:a141...	64:ff9b:6259:bac8	TCP	86	3657 → 443 [SYN] Seq=0 Win=64320 Len=0 MSS=1340 NS=256 SACK_PERM
16	0.377586700	10.244.227.10	10.244.227.127	DNS	78	Standard query 0x1cf7 A gnar.grammarly.com
17	0.378048900	10.244.227.10	10.244.227.127	DNS	78	Standard query 0x12d1 AAAA gnar.grammarly.com
18	0.411842500	10.244.227.127	10.244.227.10	DNS	206	Standard query response 0x1cf7 A gnar.grammarly.com A 44.197.81.17 A 34.198.47.121 A 52.285.199.159 A 3.2...

Observation:

A TCP packet containing the RST (Reset) and ACK flags was observed and highlighted in red. Such packets indicate that an existing connection was forcefully terminated or reset. These packets may occur because of connection interruptions, network errors, or application-level communication termination.

5. TCP THREE-WAY HANDSHAKE

TCP establishes a reliable connection using a three-step process:

SYN

The client sends a synchronization request to initiate communication.

SYN-ACK

The server acknowledges the request and responds with a synchronization acknowledgment.

ACK

The client acknowledges the server response and the connection is established.

This process ensures reliable communication between network devices.

6. RESULTS AND FINDINGS

The following observations were made during packet analysis:

- DNS packets were successfully captured and analyzed.
- Domain name resolution traffic was observed.
- HTTP requests and responses were identified.
- TCP communication packets were captured successfully.
- TCP connection control flags such as ACK and RST were observed.
- A suspicious TCP Reset packet was identified and analyzed.
- Real-time network traffic was monitored successfully using Wireshark.

7. SECURITY ANALYSIS

Network packet analysis provides visibility into network communications and helps identify unusual traffic patterns. During this project, DNS, HTTP, and TCP communications were examined successfully. The identified TCP Reset packet demonstrates how abnormal network behavior can be detected through packet inspection.

Wireshark proved to be an effective tool for monitoring network activity and troubleshooting communication issues.

8. CONCLUSION

This project successfully demonstrated the use of Wireshark for capturing and analyzing network packets. DNS, HTTP, and TCP protocols were examined in detail, and suspicious network traffic was identified through packet analysis. The project provided practical experience in network monitoring and enhanced understanding of network communication processes.

Wireshark is a powerful tool for cybersecurity professionals and network administrators for analyzing network behavior and identifying potential security concerns.

9. RESULT

The network traffic was successfully captured and analyzed using Wireshark. DNS, HTTP, TCP, and suspicious packets were identified and examined successfully. The objectives of the project were achieved.